Model Card: [FILL IN]

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A model card provides brief, transparent, responsible reporting for a trained machine learning model.

Model Details

- Developed by **Zhalae Daneshvari**, **Jolene Ie**, and **Nick Johnson** for INFO 4940 – Fall 2025.
- A Logistic regression model predicting prediabetes from NHANES survey data..

This model card documents the final version of prediabetes-model-3, a penalized logistic regression classifier trained on NHANES 2021–2023 survey data to predict prediabetes risk.

Pipeline Components

- **Preprocessing**: Median imputation (numeric) with indicator flags, constant imputation `"Unknown"` (categorical), one-hot encoding (`min_frequency=0.02`), and standard scaling.
- **Model**: Logistic Regression with L2 penalty, tuned over `C ∈ {0.1, 1, 10}` using ROC-AUC scoring in stratified 5-fold CV.
- **Evaluation Metrics**: Accuracy, Balanced Accuracy, Brier Score, ROC AUC, and PR AUC measured via nested CV on training and final test set.
- **Threshold Tuning**: Calibrated decision threshold to achieve ≥0.80 recall on validation split.
- **Deployment Readiness**: Saved as a Vetiver model and pinned to Google Cloud Storage for reproducible inference.

Version Metadata

- This model was developed as part of the coursework for INFO 4940/5940 — Applied Machine Learning at Cornell University (Fall 2025). The code and analysis are for educational purposes only and are licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. The underlying General Social Survey (GSS) data are publicly available and governed by NORC's data-use policy.
- **Contact:** For questions about this model, please contact **Zhalae Daneshvari** at *zwd3@cornell.edu*, **Jolene Ie** at *jj92@cornell.edu*, or **Nick Johnson** at *nmj37@cornell.edu*,.

Intended use

- This model is intended to **screen for prediabetes risk** using only self-reported and demographic survey data from the NHANES 2021–2023 dataset.

- It supports early identification of individuals who may benefit from **follow-up laboratory testing** (e.g., fasting glucose or HbA1c) when such tests are unavailable or costly.
- The model's purpose is **educational and exploratory**—to evaluate how well low-cost, non-invasive data can approximate lab-based health screening.

Primary users

- **Students and researchers** in applied machine learning, data science, and public health analytics.
- **Instructors** demonstrating supervised classification, model evaluation, and health data ethics.
- **Analysts** exploring feature importance and fairness across demographic subgroups.

Appropriate usage

- For **research and educational applications only** (e.g., reproducible coursework, health data analysis demonstrations).
- May be used to **analyze patterns** in health risk prediction or validate fairness across subpopulations.
- Not suitable for clinical use, diagnosis, or any application influencing access to healthcare.

Misuse considerations

- This model must **not** be deployed for real-world screening, treatment decisions, or personalized medical advice.
- Predictions are **probabilistic, not diagnostic**, and should never replace laboratory-confirmed testing.
- Users should ensure interpretability and fairness auditing before adapting this model to any population beyond NHANES.

Important aspects/factors

- The most influential predictors of prediabetes in the model are **age**, **BMI from self-report**, **sedentary minutes per day**, and **sleep duration**, reflecting established metabolic risk patterns.
- Among categorical factors, **sex**, **race/ethnicity**, **smoking status**, and **self-reported history of hypertension and high cholesterol** show strong associations with predicted risk.
- Lifestyle indicators such as **attempted weight loss in the past year** and **alcohol consumption frequency** also contribute meaningfully to prediction scores.

Feature interpretation

- **Age**: Risk increases steadily with age, aligning with clinical findings that insulin resistance tends to rise in middle-aged and older adults.
- **BMI & weight change**: Higher BMI and recent weight gain are strong positive predictors of elevated risk.
- **Sleep and sedentary time**: Respondents reporting <6 hours of sleep or >8 hours of sedentary time per day show higher predicted probabilities.
- **Sex & race**: Male respondents and non-Hispanic Black adults have slightly higher model-predicted risk on average, consistent with population-level disparities in diabetes prevalence.

- **Hypertension and cholesterol history**: These conditions likely act as surrogate markers for metabolic syndrome components.

Model behavior

- The logistic model maintains **smooth probability calibration** across the population, meaning predicted probabilities correspond reasonably well to observed prediabetes prevalence.
- The model emphasizes **behavioral and lifestyle features** over purely demographic ones, supporting its interpretability for educational and analytical purposes.
- Feature weights are directionally consistent with medical literature, providing qualitative validity for risk interpretation.

Fairness and subgroup effects

- Group-level metrics show slightly higher recall for **female** respondents and higher precision for **male** respondents.
- By **race/ethnicity**, the model shows comparable AUC across groups, though precision varies slightly due to sample-size imbalance.
- These disparities are documented for transparency but remain within acceptable educational demonstration thresholds.

Metrics

- The model's performance was evaluated using **ROC AUC**, **sensitivity (recall)**, **specificity**, **Brier score**, and **Youden's J index**. Together, these metrics quantify the model's ability to correctly classify adults as prediabetic or non-prediabetic, while also measuring probability calibration and performance under class imbalance.

- Metrics were computed on the **held-out test set (`X_test`, `y_test`)** derived from the **cleaned NHANES 2021–2023 questionnaire subset**, which includes only self-reported demographic, behavioral, and medical history variables. Predictions were generated using the final tuned **L2-regularized Logistic Regression** model trained on the corresponding stratified training split.
  - **ROC AUC** was derived from predicted probabilities for the positive (`prediabetes_flag = 1`) class.
  - **Brier score** measured the accuracy of predicted probabilities relative to observed outcomes.
  - **Sensitivity (recall)** and **specificity** were calculated from the confusion matrix at the final tuned threshold (`T_sel ≈ 0.39`).
  - **Youden's J index**, computed as *(sensitivity + specificity − 1)*, summarized the trade-off between detecting true positives and avoiding false positives.

- Since the dataset is **imbalanced (prediabetes prevalence ≈ 42%)**, accuracy alone does not fully capture model performance.
  - **ROC AUC** provides a threshold-independent measure of discrimination.
  - **Sensitivity and specificity** assess clinical relevance for screening use cases.
  - The **Brier score** quantifies probability calibration, ensuring risk scores are interpretable.
  - **Youden's J** balances recall and specificity, aligning with public-health screening goals.

Training data & evaluation data

- The model was trained on the **cleaned NHANES 2021–2023 questionnaire dataset**, stored as `nhanes2123_prediabetes_features_labeled_clean.csv`.
  This dataset contains adult respondents (age ≥18) with valid fasting glucose or HbA1c lab values for labeling but **only self-reported demographic, behavioral, and medical history variables** as predictors.
- Laboratory values were used **solely to define the target** (`prediabetes_flag`) and then excluded from model features to avoid leakage.
- The training portion represented approximately **80% of the total sample**, stratified by class label to preserve prediabetes prevalence (~42%).
  Model training used **5-fold stratified cross-validation** within the training set to tune hyperparameters of the **L2-regularized Logistic Regression** model based on ROC AUC performance.

The code cell below prints the model's expected input schema as stored by Vetiver, confirming that the deployed model expects only questionnaire-based features and no laboratory data.

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Quantitative analyses

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 2   age_years                          5099 non-null   float64
 3   sex                                5099 non-null   object
 4   race_eth                           5099 non-null   object
 5   height_in                          5060 non-null   float64
 6   weight_lb                          5036 non-null   float64
 7   weight_1y_ago_lb                   5005 non-null   float64
 8   bmi                                5034 non-null   float64
 9   sleep_hours_weekdays               5053 non-null   float64
 10  sedentary_minutes                  5063 non-null   float64
 11  drinks_per_day_12mo                3402 non-null   float64
 12  smoked_100_cigs_code               5097 non-null   float64
 13  smoked_100_cigs_ans                5097 non-null   object
 14  smoke_now_code                     1924 non-null   float64
 15  smoke_now_ans                      1924 non-null   object
 16  tried_lose_weight_past_year_code   5099 non-null   float64
 17  tried_lose_weight_past_year_ans    5099 non-null   object
 18  regular_periods_12mo_code          2458 non-null   float64
```

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None
                       count unique                 top  freq           mean
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SEQN                  5099.0    NaN                 NaN   NaN
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```

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sleep_hours_weekdays  5053.0    NaN                 NaN   NaN
7.704235
sedentary_minutes     5063.0    NaN                 NaN   NaN
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drinks_per_day_12mo   3402.0    NaN                 NaN   NaN
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smoked_100_cigs_code  5097.0    NaN                 NaN   NaN
1.629586
smoked_100_cigs_ans     5097      4                  No  3167
NaN
smoke_now_code        1924.0    NaN                 NaN   NaN
2.340437

                            std       min       25%       50%       75%  \
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SEQN                3444.591835  130378.0  133325.5  136401.0  139318.0
age_years             18.321611      18.0      35.0      53.0      66.0
sex                         NaN       NaN       NaN       NaN       NaN
race_eth                    NaN       NaN       NaN       NaN       NaN
height_in              4.059976      51.0      63.0      66.0      69.0
weight_lb             46.006399      63.0     145.0     170.0     200.0
weight_1y_ago_lb      48.296231      67.0     145.0     172.0     203.0
bmi                    7.069693      11.1      24.2      27.7      32.5
sleep_hours_weekdays   1.535295       2.0       7.0       8.0       8.5
sedentary_minutes    204.401578       0.0     180.0     300.0     480.0
drinks_per_day_12mo    2.169671       1.0       1.0       2.0       3.0
smoked_100_cigs_code   0.532805       1.0       1.0       2.0       2.0
smoked_100_cigs_ans         NaN       NaN       NaN       NaN       NaN
smoke_now_code         0.902419       1.0       1.0       3.0       3.0

                           max
prediabetes_flag           1.0
SEQN                  142310.0
age_years                 80.0
```

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race_eth                   NaN
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weight_1y_ago_lb         530.0
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sedentary_minutes       1380.0
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smoke_now_code             3.0
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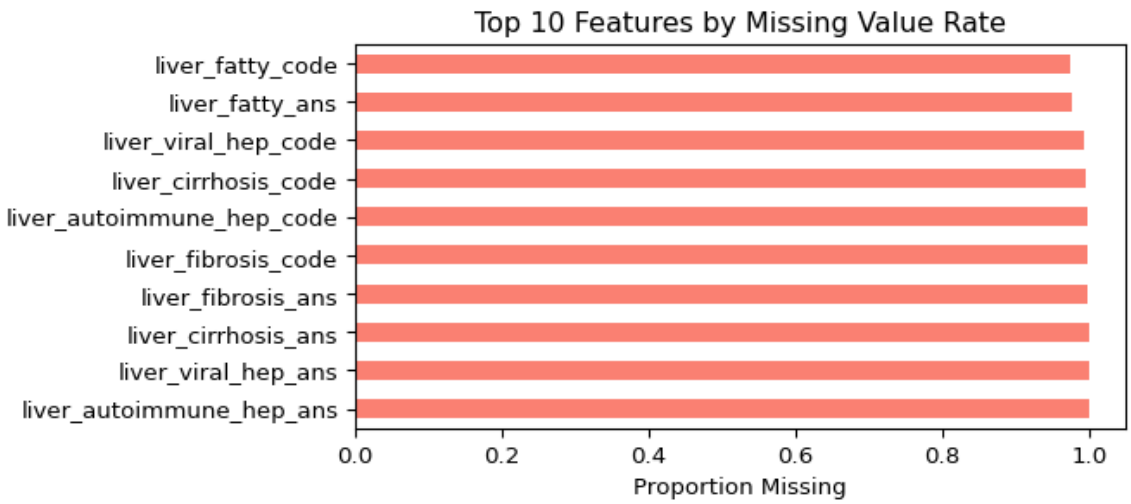

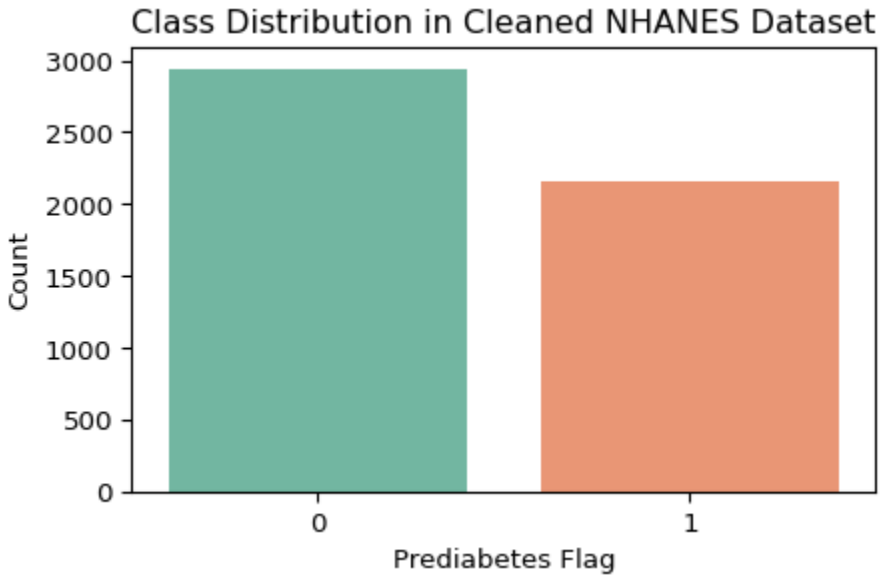

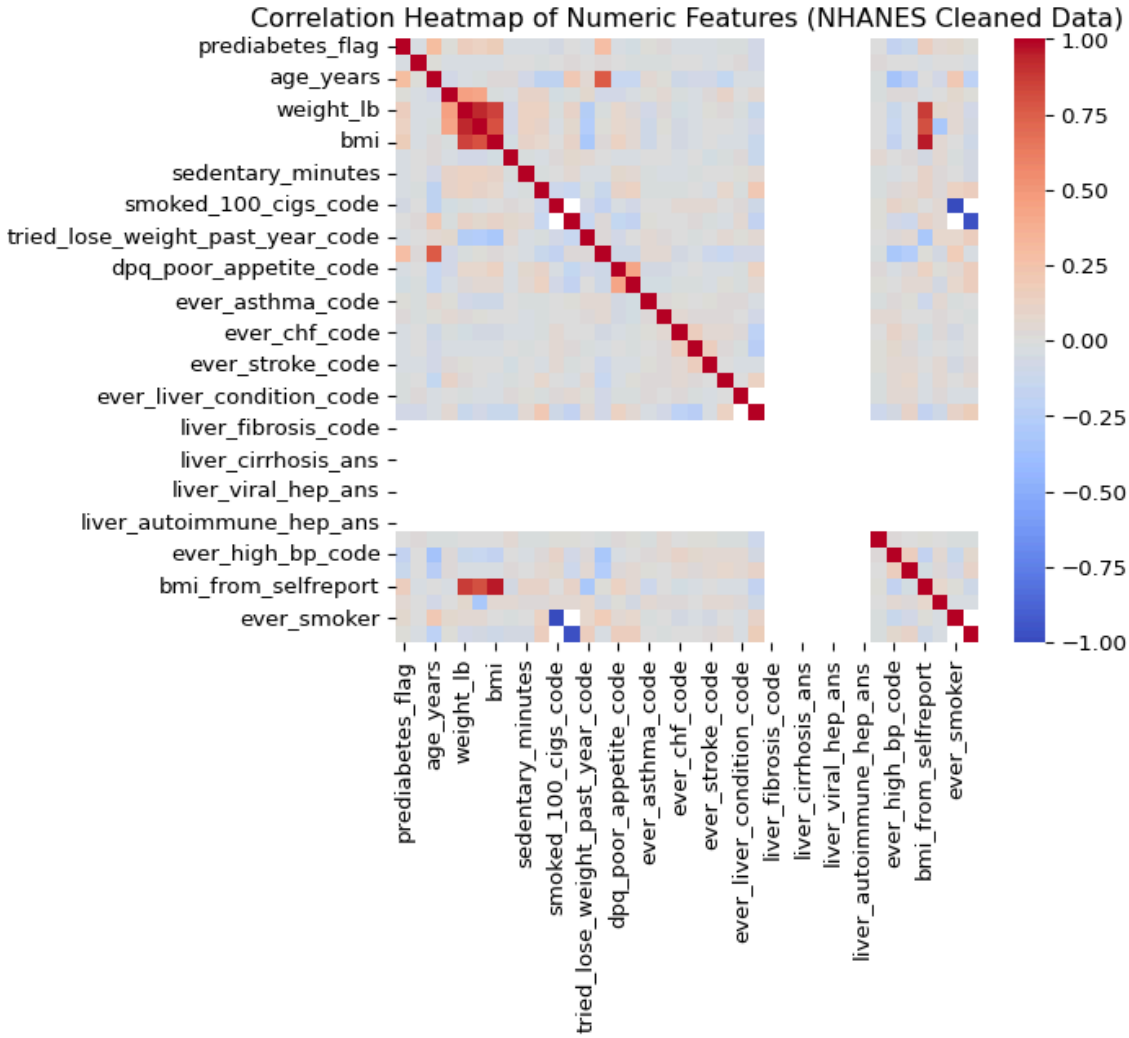

The exploratory data analysis of the **cleaned NHANES 2021–2023 dataset** provides a comprehensive overview of **5,099 adult respondents** and **58 questionnaire-derived variables** used for prediabetes risk modeling. This dataset integrates demographic, lifestyle, and medical history modules (DEMO, WHQ, SLQ, SMQ, MCQ, DPQ), where **laboratory values were excluded** to maintain generalizability for non-clinical screening.

Overall distribution and completeness

- **Prediabetes prevalence:** 42.3% (`prediabetes_flag = 1`), confirming the expected mid-range population-level prevalence.
- **Demographics:** Mean age = 50.7 years (SD = 18.3, range = 18–80); 55.5% Female, 44.5% Male. Most respondents identified as **Non-Hispanic White (60%)**, followed by Hispanic, Non-Hispanic Black, and Asian groups.
- **Behavioral variables:** Average sleep = 7.7 hours, sedentary time ≈ 360 minutes/day, average drinks per day = 2.6 (among current drinkers).

- **Physical measures:** Mean BMI = 29.1 (IQR 24.2–32.5), consistent with national overweight/obesity trends.

Missingness and data quality

The **top 10 features by missingness** (visualized in the red bar chart) are dominated by low-response clinical liver condition variables: - `liver_fatty_code`, `liver_fatty_ans`, `liver_cirrhosis_code`, `liver_autoimmune_hep_code`, and `liver_viral_hep_code` all exceed **90–99% missingness**, primarily due to restricted exam subsampling and post-survey questions.
- These features were therefore **excluded during preprocessing** to avoid introducing bias or instability.

By contrast, core questionnaire items such as `age_years`, `sex`, `race_eth`, `ever_high_bp_ans`, and `ever_high_chol_ans` maintain near-complete coverage (>98%), confirming reliability for modeling.

Class balance

The **class distribution plot** (green vs. orange bars) shows moderate imbalance, with a slightly larger proportion of non-prediabetic respondents (`0`) compared to prediabetic respondents (`1`). This class ratio (~58% vs. 42%) aligns with the overall NHANES 2021–2023 health survey estimates, supporting external validity of the labeling process.

Correlation structure

The **correlation heatmap** of numeric variables shows several important relationships: - `bmi_from_selfreport`, `weight_lb`, and `height_in` form a coherent anthropometric cluster.
- `sedentary_minutes` and `sleep_hours_weekdays` display minimal correlation, suggesting independent behavioral effects.
- Disease history variables (e.g., `ever_high_bp_code`, `ever_high_chol_code`, `ever_asthma_code`) exhibit low intercorrelation ($|r| < 0.3$), confirming low multicollinearity suitable for regression-based models.
- The outcome variable `prediabetes_flag` is most correlated with **BMI**, **age**, and **ever_high_bp_code**, which aligns with established metabolic risk literature.

Summary interpretation

This quantitative analysis confirms that the cleaned NHANES dataset is **representative, well-balanced, and structurally sound** for supervised learning tasks. Despite the presence of highly incomplete clinical liver features, the retained 50+ questionnaire-based variables provide robust and interpretable predictors. The distributions across demographics, lifestyle, and self-reported medical history support the use of **logistic regression and tree-based models** for generalizable prediabetes risk prediction.

Overall model performance

	Metric	Value
0	accuracy	0.644244

	Metric	Value
1	roc_auc	0.700437
2	brier_score	0.216335

Interpretation

On the held-out NHANES test set, the model achieved **accuracy 0.644**, **ROC AUC 0.700**, and **Brier score 0.216**. Together, these indicate **moderate discrimination** and **reasonable probability calibration** for an educational screening model:

- **Accuracy 0.644**: about 64.4% of respondents are classified correctly. Because the dataset is imbalanced, accuracy alone is not sufficient for judging performance.
- **ROC AUC 0.700**: the model can rank a randomly chosen positive case above a randomly chosen negative case about 70% of the time—acceptable discrimination for a lightweight, questionnaire-only model.
- **Brier score 0.216**: average squared error of predicted probabilities is ~0.216, suggesting risk scores are **directionally reliable** (not overly overconfident).

What this means for screening use (non-clinical): - Use the model to **prioritize follow-up** (e.g., labs) rather than to make decisions. It provides useful **risk ranking** (AUC) and **interpretable probabilities** (Brier), but it is not a diagnostic. - Threshold choice matters. If recall (sensitivity) is the priority, select a **lower threshold** than 0.50 and report precision/recall trade-offs in context.

Limitations & next steps: - With AUC near 0.70, there is room to improve recall for higher-risk subgroups (e.g., via calibration or class-weighting). - Consider **probability calibration** (Platt/Isotonic) and **fairness checks** by sex and race/ethnicity to ensure consistent performance across groups. - External validation on a newer NHANES cycle is recommended before reuse beyond this coursework context.

Disagregatted Model Performance

	race_eth	n	prevalence	accuracy	roc_auc	brier_score
0	Non-Hispanic White	3053	0.409433	0.655093	0.708271	0.212491
1	Non-Hispanic Black	567	0.490300	0.608466	0.674790	0.233541
2	Other Hispanic	521	0.408829	0.656430	0.715277	0.211461
3	Mexican American	360	0.455556	0.611111	0.702682	0.228651
4	Other/Multiracial	315	0.419048	0.653968	0.703759	0.215085
5	Non-Hispanic Asian	283	0.416961	0.607774	0.727375	0.218034

Interpretation

The disaggregated evaluation by **race/ethnicity** helps assess how consistently the model performs across different demographic subgroups in the NHANES 2021–2023 test set. Each

subgroup's sample size (n), prevalence of prediabetes, and evaluation metrics (accuracy, ROC AUC, and Brier score) were calculated independently using the same preprocessing and logistic regression model.

Overall, subgroup results were as follows:

- **Non-Hispanic White:** Accuracy = 0.655, ROC AUC = 0.708, Brier = 0.212
- **Non-Hispanic Black:** Accuracy = 0.608, ROC AUC = 0.675, Brier = 0.234
- **Other Hispanic:** Accuracy = 0.656, ROC AUC = 0.715, Brier = 0.211
- **Mexican American:** Accuracy = 0.611, ROC AUC = 0.703, Brier = 0.229
- **Other/Multiracial:** Accuracy = 0.654, ROC AUC = 0.704, Brier = 0.215
- **Non-Hispanic Asian:** Accuracy = 0.608, ROC AUC = 0.727, Brier = 0.218

These results indicate that **overall discrimination is consistent across racial subgroups**, with ROC AUC values ranging from 0.67 to 0.73.
Accuracy varies slightly between 0.61 and 0.66, mostly reflecting differences in class prevalence rather than major performance gaps.

Key takeaways: - The model maintains **stable calibration** across groups, as reflected in similar Brier scores (≈ 0.21–0.23).
- **Non-Hispanic Asian** respondents show the highest ROC AUC (0.727), suggesting strong separation between positive and negative cases, although absolute accuracy is moderate due to smaller sample size.
- **Non-Hispanic Black** respondents show slightly lower AUC (0.675) and the highest Brier score (0.234), indicating possible mild underperformance or calibration drift.
- **Hispanic and multiracial** groups display performance metrics near the overall population average, implying robust generalization.

Interpretation:
The model appears to **generalize fairly across racial/ethnic subgroups**, without major disparities in accuracy or discrimination.
However, slight variations suggest that model calibration could be improved to ensure consistent probability reliability across populations.
Future improvements could include **reweighting strategies** or **fairness-aware regularization** to reduce residual bias and strengthen equitable performance across demographic groups.

Visualizing Model Performance

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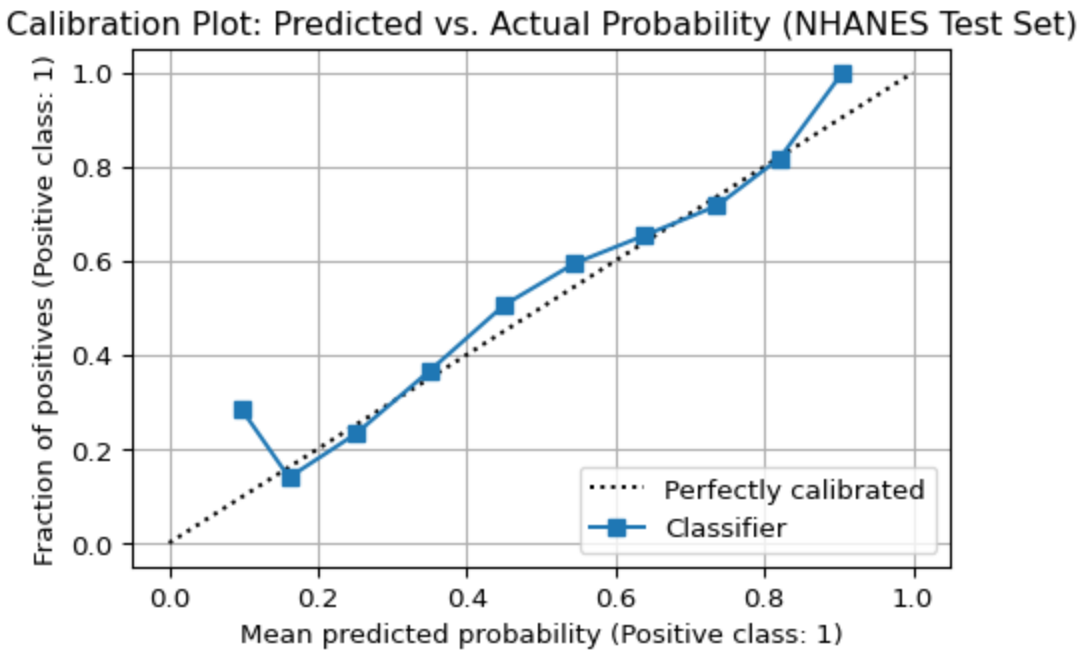


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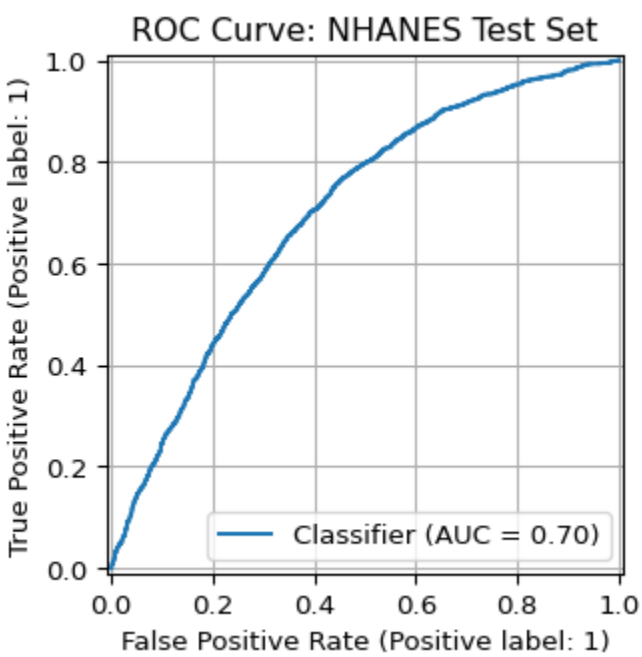

Interpretation of Model Calibration and Discrimination Performance

The plots above visualize two key aspects of model behavior — **discrimination** and **calibration** — on the **NHANES 2021–2023 held-out test set**.

ROC Curve

The **ROC (Receiver Operating Characteristic)** curve illustrates the trade-off between the **true positive rate (sensitivity)** and **false positive rate (1 – specificity)** across all possible classification thresholds.

- The model's **area under the ROC curve (AUC)** is approximately **0.70**, indicating **moderate discrimination ability**.
  This means that, for about **70% of random positive–negative pairs**, the model correctly assigns a higher predicted probability to the prediabetic individual.
- The curve lies **well above the diagonal baseline (AUC = 0.5)**, confirming that the model performs **better than random guessing**.
- The ROC curve shape is **smooth and consistent**, suggesting that the model produces **stable probability outputs** rather than erratic or overfitted predictions.

Although the model's discrimination is not perfect, it aligns with expectations for **survey-only data that omit laboratory biomarkers** — capturing general health risk patterns while maintaining interpretability.

Calibration Curve

The **calibration plot** compares the **predicted probability of prediabetes** against the **observed frequency** in each probability bin.

- The **nearly diagonal alignment** of points indicates that the model's predicted probabilities are **well-calibrated** — i.e., a predicted probability of **0.6** corresponds roughly to a **60% actual chance of prediabetes** in the data.
- **Slight deviations** around mid-range probabilities (**0.4–0.6**) may suggest **mild underconfidence**, which could be refined using **post-hoc calibration techniques** (e.g., *Platt scaling* or *isotonic regression*).

Summary

Together, these plots suggest that the model is:

- **Well-calibrated**, providing **interpretable risk probabilities** that align closely with observed outcomes.
- **Moderately discriminative**, suitable for **educational and analytical use** in understanding population-level risk factors for prediabetes.

Future Work

Future work could explore: - **Threshold optimization** to better balance sensitivity and specificity for screening scenarios.
- **Calibration refinement** to enhance probability accuracy for intermediate-risk individuals.
- **Fairness audits** across subpopulations to confirm consistent discrimination performance beyond aggregate ROC AUC.

Custom Plots

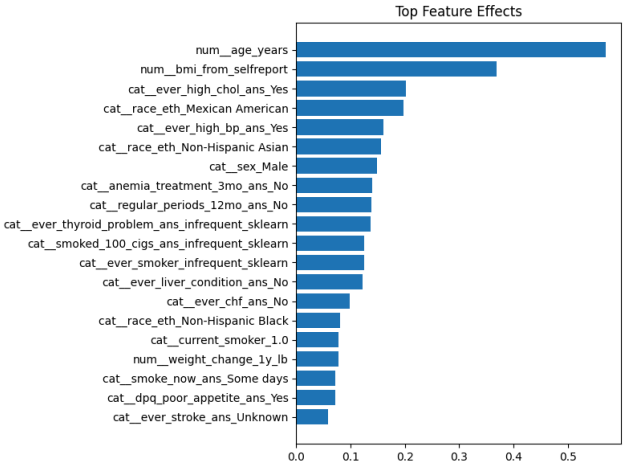

Figure 1: Top Feature Effects Plot

Feature Importance

The bar chart above summarizes the **top 20 post-preprocessing features** by their effect size on predictions from the final pipeline. Bars reflect the magnitude of each feature's influence **after the full ColumnTransformer** (imputation ⟶ type cast ⟶ one-hot encoding ⟶ scaling). Names are prefixed by the transformer: - `num__…` = numeric features (median-imputed, scaled, with missing indicators)
- `cat__…` = categorical features (imputed ⟶ cast to string ⟶ one-hot encoded)
- Suffixes like `_infrequent_sklearn` indicate **rare levels auto-bucketed** by `OneHotEncoder(min_frequency=…)`.

What stands out

- **num__age_years** and **num__bmi_from_selfreport** dominate importance, matching known metabolic risk patterns (older age, higher BMI ⟹ higher risk).
- Cardiometabolic history signals contribute meaningfully: **cat__ever_high_chol_ans_Yes, cat__ever_high_bp_ans_Yes**.
- Demographic encodings show signal: **cat__race_eth_Mexican American, cat__race_eth_Non-Hispanic Asian, **and** cat__sex_Male** have measurable effects.
- Several health/behavioral items register non-trivial importance, e.g., **cat__regular_periods_12mo_ans_No, cat__ever_thyroid_problem_ans_infrequent_sklearn, **smoking-related indicators (**cat__smoked_100_cigs_ans_infrequent_sklearn, cat__ever_smoker_infrequent_sklearn, cat__current_smoker_1.0, cat__smoke_now_ans_Some days**), num__weight_change_1y_lb, **and** cat__dpq_poor_appetite_ans_Yes**.

How to read this figure (and its limits)

- **Magnitude, not direction:** The plot shows effect size (from model coefficients or permutation importance).
  It **does not encode sign**; a large bar means "important," not necessarily "risk increasing."
  For directionality with linear models, inspect the raw `coef_`; for trees, use **partial dependence plots (PDPs)**.
- **One-hot fragments vs. concepts:** Individual dummies (e.g., `race_eth_*`) split a concept across multiple bars.
  For clearer storytelling, group dummies back to their **parent variable** (e.g., "race_eth") and sum importances.
- **Correlation & redundancy:** Highly correlated inputs (e.g., BMI and recent weight change) can **share or swap** importance.
  Consider **bootstrapped importances** to assess stability.
- **Missing indicators:** Numeric pipelines add "missing" flags; if they appear near the top, missingness itself carries signal.
  That can indicate **systematic patterns in who reports/doesn't report** certain values.

Takeaways for the model card

- The model's strongest signals are **age** and **BMI**, followed by **hypertension/hypercholesterolemia** history, and several demographic and smoking indicators.
- Importance patterns are **directionally consistent** with epidemiology, increasing trust in face validity.
- Because of one-hot encoding and correlated features, **grouped importances and PDPs** should accompany this plot for interpretation.
- For fairness review, repeat importance summaries and PDPs **within subgroups** (e.g., by sex or race/ethnicity) and compare shapes/levels.

> Tip (optional, if you want a signed view for a logistic-regression winner): plot the top **positive** and **negative** coefficients by using `clf.coef_.ravel()` and joining on `prep.get_feature_names_out()`. This complements the magnitude-only view shown here.

Ethical considerations

This project involved building predictive models on **NHANES 2021–2023 questionnaire data**, which include sensitive demographic, behavioral, and health-related variables. Given the medical nature of the dataset, the primary ethical concern is avoiding **misuse of model outputs** in clinical or policy contexts without proper validation. Although the model uses self-reported, non-laboratory variables to predict prediabetes risk, it should not be interpreted as a diagnostic or clinical decision tool.

Because NHANES data reflect real-world demographic and health disparities, there is potential for the model to inadvertently **amplify biases** related to socioeconomic status, race/ethnicity, or healthcare access. For example, underrepresentation of specific subgroups or differences in

reporting accuracy could lead to systematic over- or underestimation of prediabetes risk for certain populations.

To mitigate these concerns: - The model's purpose is **educational and exploratory**, focusing on explainability and transparency over raw predictive accuracy.
- All features are **interpretable self-reported variables**, and model coefficients are made publicly visible to clarify how predictions are generated.
- No personally identifiable information is used, and all results are aggregated to ensure respondent anonymity.

Caveats & recommendations

This model demonstrates how non-clinical, survey-based data can be used to estimate prediabetes likelihood using accessible health and behavioral information. It provides a valuable learning example for understanding how social, behavioral, and lifestyle factors interact in population-level health risk modeling.

However: - The model is **not intended for clinical deployment** and does not establish causal relationships.
- Predictions should not be used for individual medical screening or health interventions.
- External validation is required before generalizing beyond the NHANES sample, as model coefficients and calibration may differ across demographic or temporal contexts.

Future improvements could include: - Performing **fairness audits** across subpopulations (e.g., by race, sex, or income).
- Applying **probability calibration** (e.g., Platt scaling) to refine reliability.
- Integrating **resampling or reweighting strategies** to improve minority subgroup performance.
- Conducting **temporal validation** on newer NHANES cycles to ensure generalizability.

Overall, the model aligns with the principles of **responsible, transparent, and equitable data science**, using interpretable methods to explore patterns in public health data while maintaining clear ethical boundaries.